Paper Review • arXiv 2611.12806 • cs.RO

LingBot-Touch: One Encoder for Three Tactile Sensor Types

Learning Heterogeneous Tactile Representation for General Dexterous Manipulation

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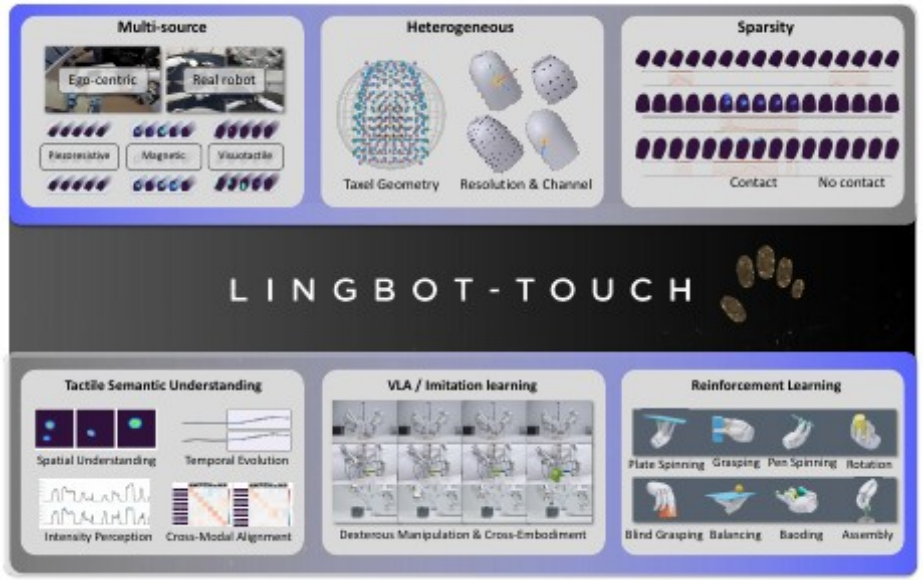
Ant Group • Robbyant First submitted 2026-11-09

Piezoresistive • Magnetic • Visuotactile | Tactile-JEPA | Plug-in tokens



Source: paper Fig.1 (LingBot-Touch overview)

TL;DR

Frozen 5M-param tactile encoder lifts four policy types by 14–25 pp

Unified coordinates + latent prediction → general tactile representation.

Maps piezoresistive, magnetic and visuotactile taxels into a canonical fingertip-sphere frame and pretrains a shared Haptic Transformer with Tactile-JEPA on ~5,000 h of glove + teleoperation data; frozen, it plugs into IL / VLA / RL policies as tokens and transfers to unseen sensor models with zero parameter updates.

4,983 h

Pretraining corpus
3 sensing principles

+13.9~24.8

26 real-robot tasks
avg. gain (pp)

69.0%

12 sim RL tasks
only 21.4% w/o touch

0 param updates

Transfer to unseen XHand1 magnetic sensor

<0.2%

Tiny encoder params as
share of host policy

Note: real-robot gain is the mean over 26 tasks (Fig.11 / Tab.2); RL is the equal-weight mean over 12 tasks (Tab.3).

Motivation

Why touch fails to scale: data are tied to sensor physics

① No unified input format

Sensors differ in channels, layout, resolution and sampling rate; one sensor's readings are meaningless to a model trained on another.

② No large-scale corpus

Data cannot be pooled across sensors; every hour of data requires a sensorized hand physically touching the world.

③ No established recipe

Evidence that “touch helps” is scattered across isolated per-sensor, per-task studies, lacking a comparable shared representation.

Key Insight

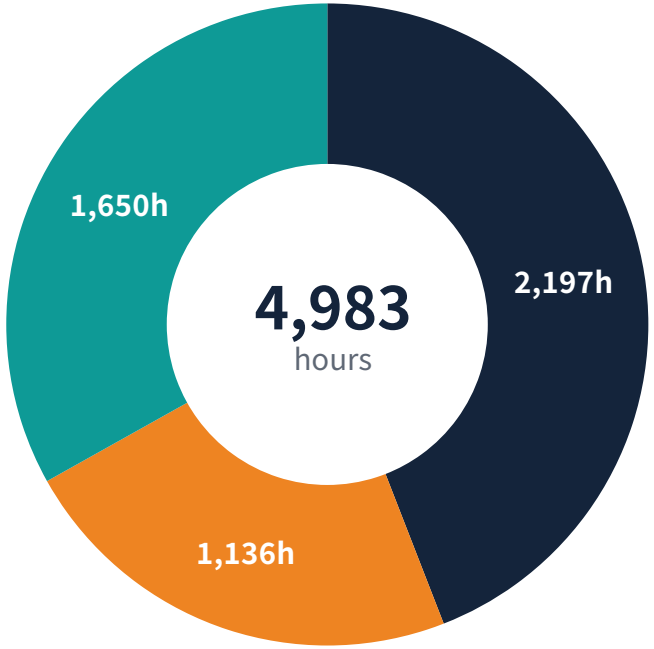
Contact itself is sensor-agnostic

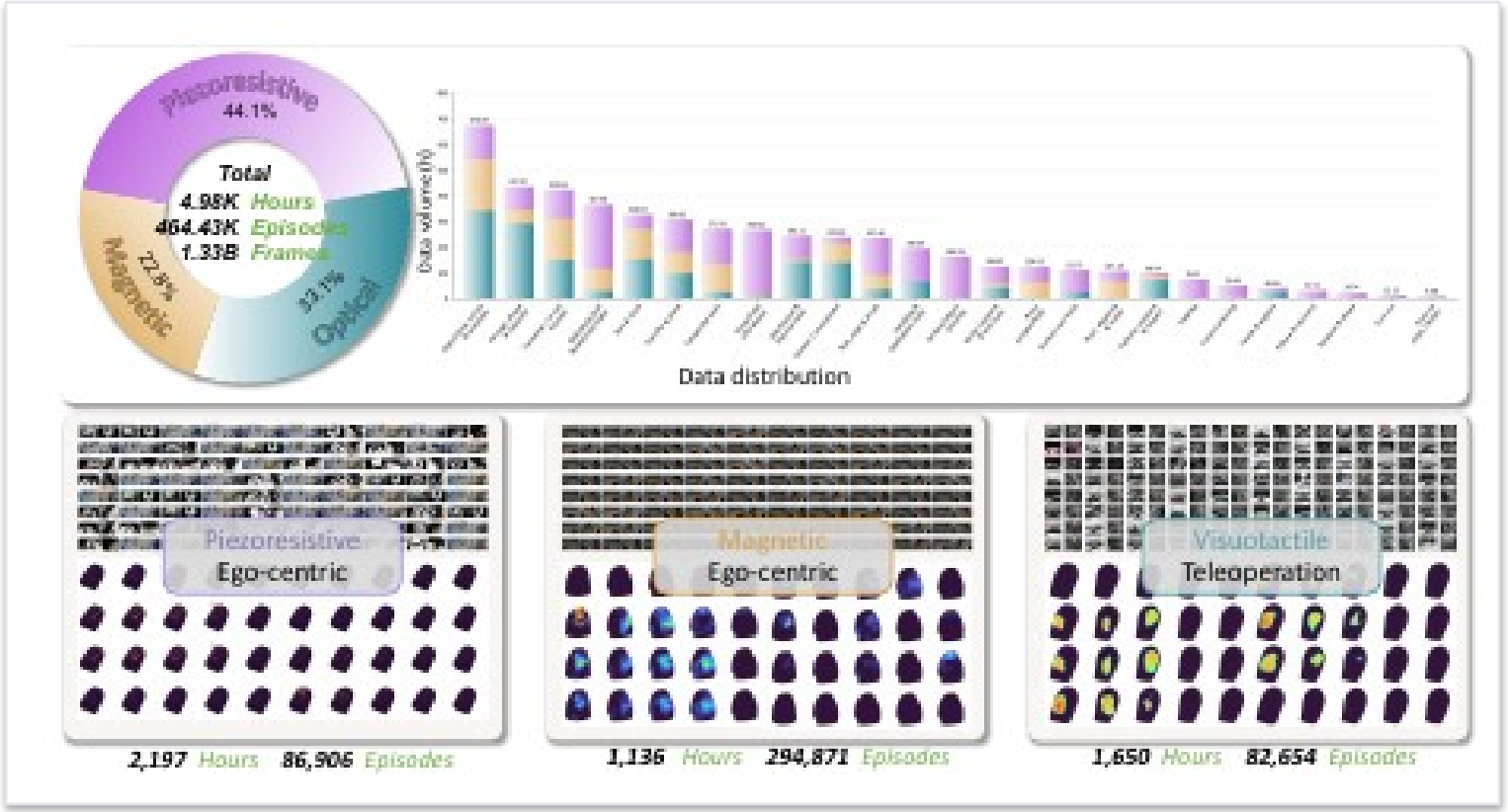
A fingertip “pressing on an edge” is the same physical event whether recorded by a magnetometer, a resistive grid or a gel camera.

→ Strip away sensor-specific physics, and contact representations can be shared.

Data

5,000 h corpus: ⅔ from human gloves, ⅓ from robot teleoperation





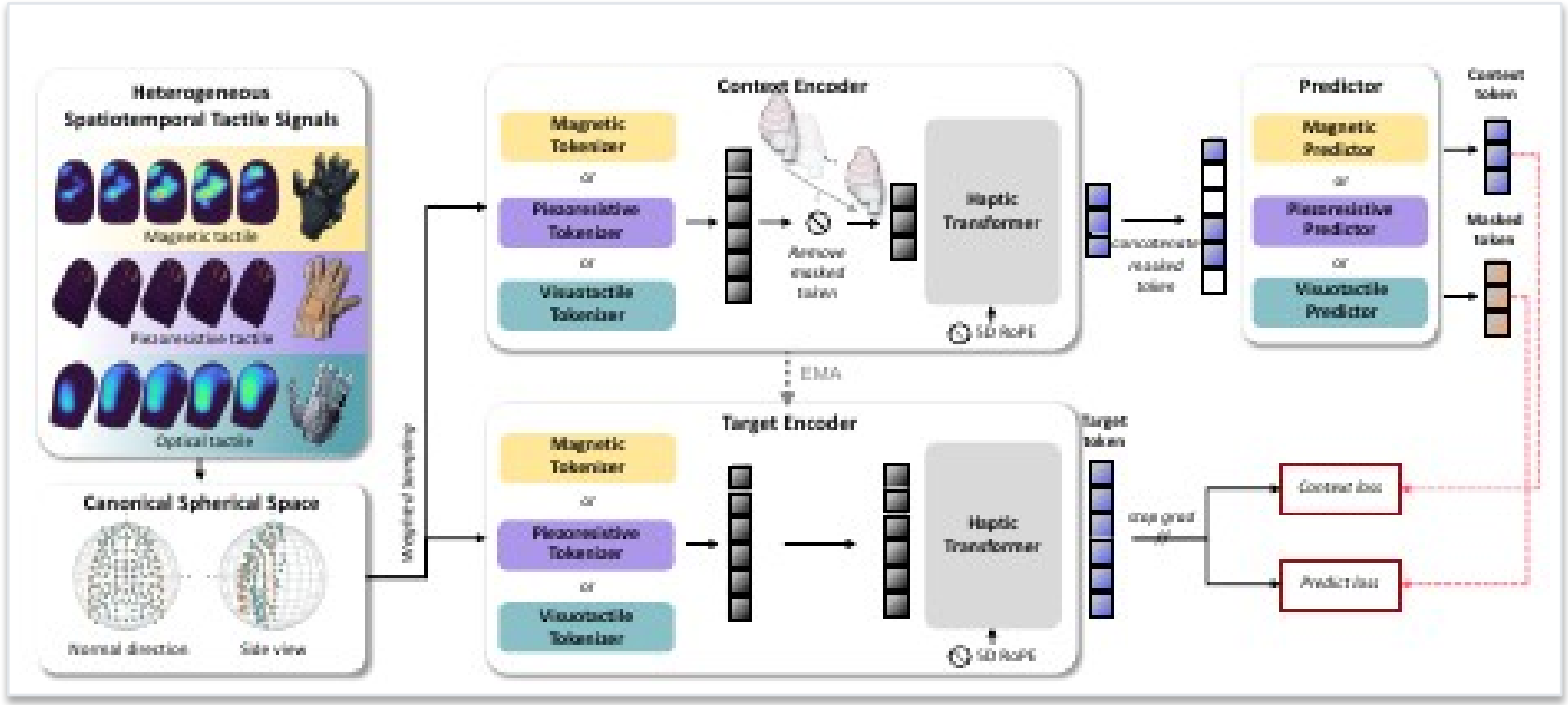
464K
episodes

1.33B
frames

24
task types, most multi-source

Class labels only
No force/pos./time GT → SSL

Tactile-JEPA: predict masked taxels in latent space, not raw signals



① Unified input

Taxels → canonical fingertip coords; 5D addr. (t, finger, x, y, z) only via 5D RoPE; values range-normalized.

② Specific ends, shared core

Mag./piezo./visuotactile: 2-layer MLP tokenizers (token = 1 taxel × 2 frames × 3 ch.); shared 12-layer HaT.

③ Latent target

Target: 4 concat. EMA-encoder depth features; L1 loss: masked + 0.5 × visible; frozen EMA encoder downstream.

Four key designs adapt JEPA to sparse, heterogeneous tactile signals

Canonical fingertip space

Each sensor's fingertip geometry is scaled and shifted via its minimal enclosing sphere onto a unit reference sphere: $p = \alpha\tilde{p} + b$.

The model sees "positions on the finger", not sensor array indices.

5D RoPE positional encoding

Each axis of the address (time window, finger, 3D coords) takes a slice of the attention head dim.

No fixed token count or sensor-specific position table → new layouts plug in directly.

Masking-ratio warm-up

The same taxels are masked over the whole clip; final mask ratio 80%.

P(sparse contact fully masked) $\approx 0.8^{k}$ → early gradient explosion; 40%→80% over first 300 steps (200K total).

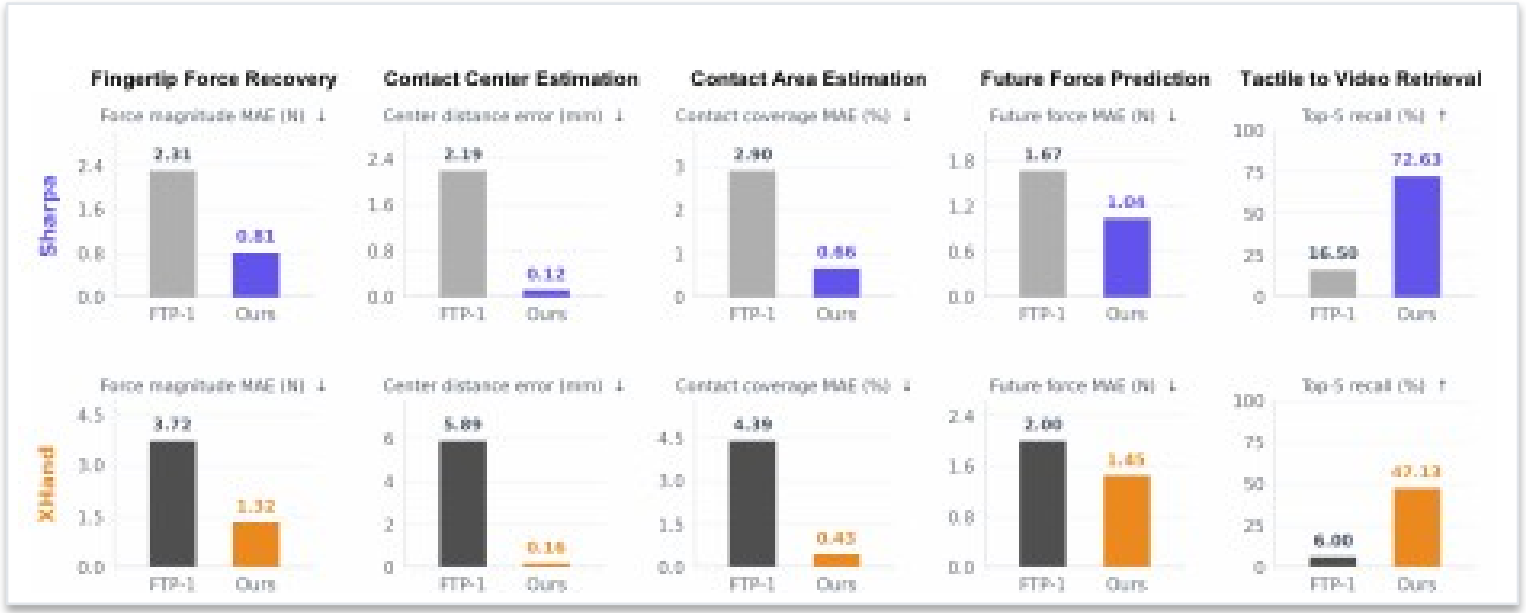
Layout-agnostic augmentation

Multi-resolution: magnetic 24–96 taxels/finger, visuotactile 32/96/200 virtual taxels.

Position jitter: $\|\delta_i\| \leqslant 0.5 \cdot d_i$ (nearest-neighbor distance), shared by both branches.

Model sizes Tiny: 5.46M params (width 192) Small: 21.76M params (width 384) both 12-layer Transformers, 200,000 pretraining steps

Frozen features read contact; far beyond FTP-1, new sensors too



0.16 mm

XHand1 (unseen) contact-center error
FTP-1: 5.89 mm

47.1% vs 6.0%

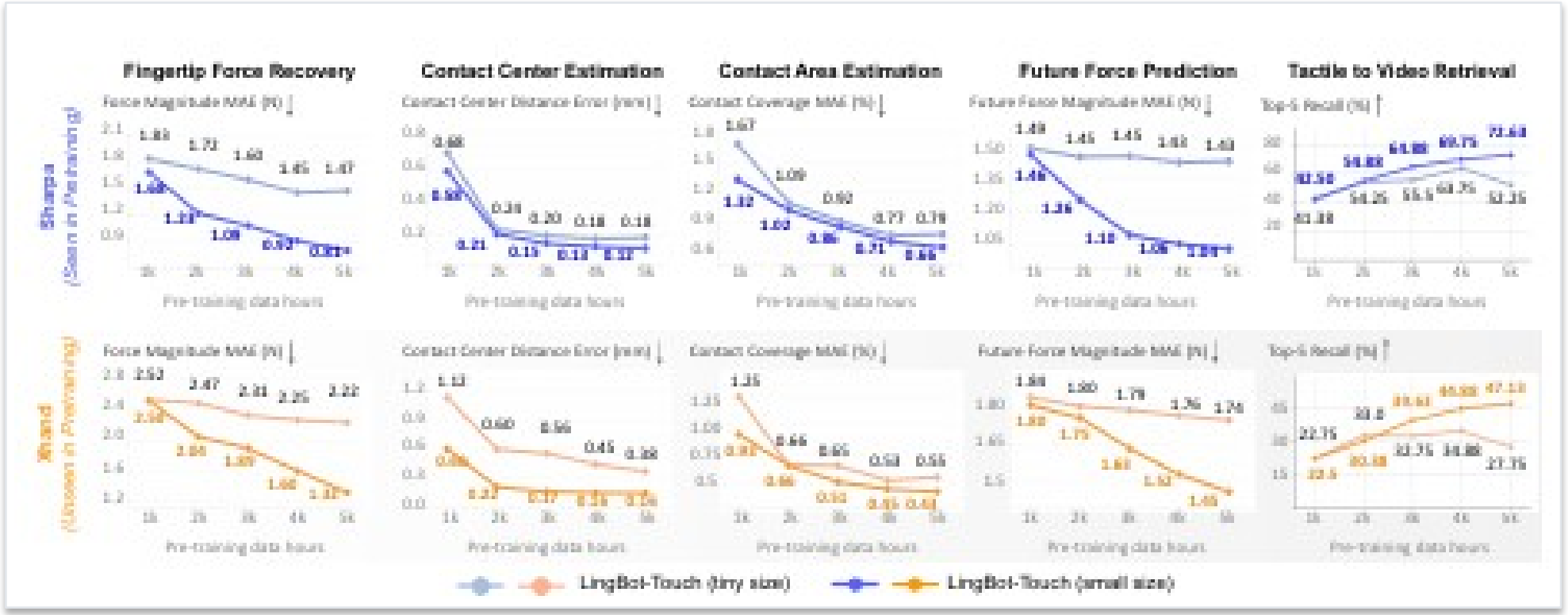
XHand1 tactile→video retrieval R@5

0.81 N vs 2.31 N

Sharpa fingertip force recovery MAE
(range 0–30 N)

Protocol: frozen EMA target encoder + one attentive probe per task; FTP-1 freezes its backbone but trains a dedicated adapter per sensor / task. Gray: FTP-1; color: Ours-Small (5K h).

More data yields better representations, but capacity must grow in step



Sharpa (seen) • Small

Force MAE 1.68 → 0.81 N
Retrieval R@5 42.5% → 72.6%

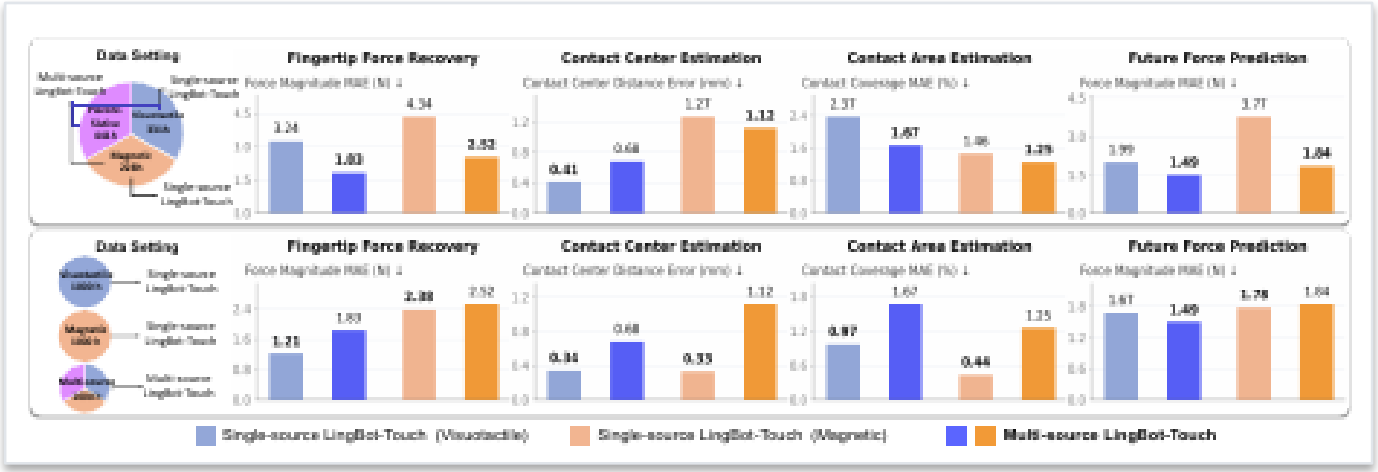
XHand1 (unseen) • Small

Force MAE 2.50 → 1.32 N
Retrieval R@5 22.5% → 47.1%

Tiny's ceiling

Flat after ~4K h; R@5 falls on both hands at 5K h → capacity must scale

Force/dynamics transfer; spatial precision is layout-specific



Own data fixed + other sources → 7/8 improve

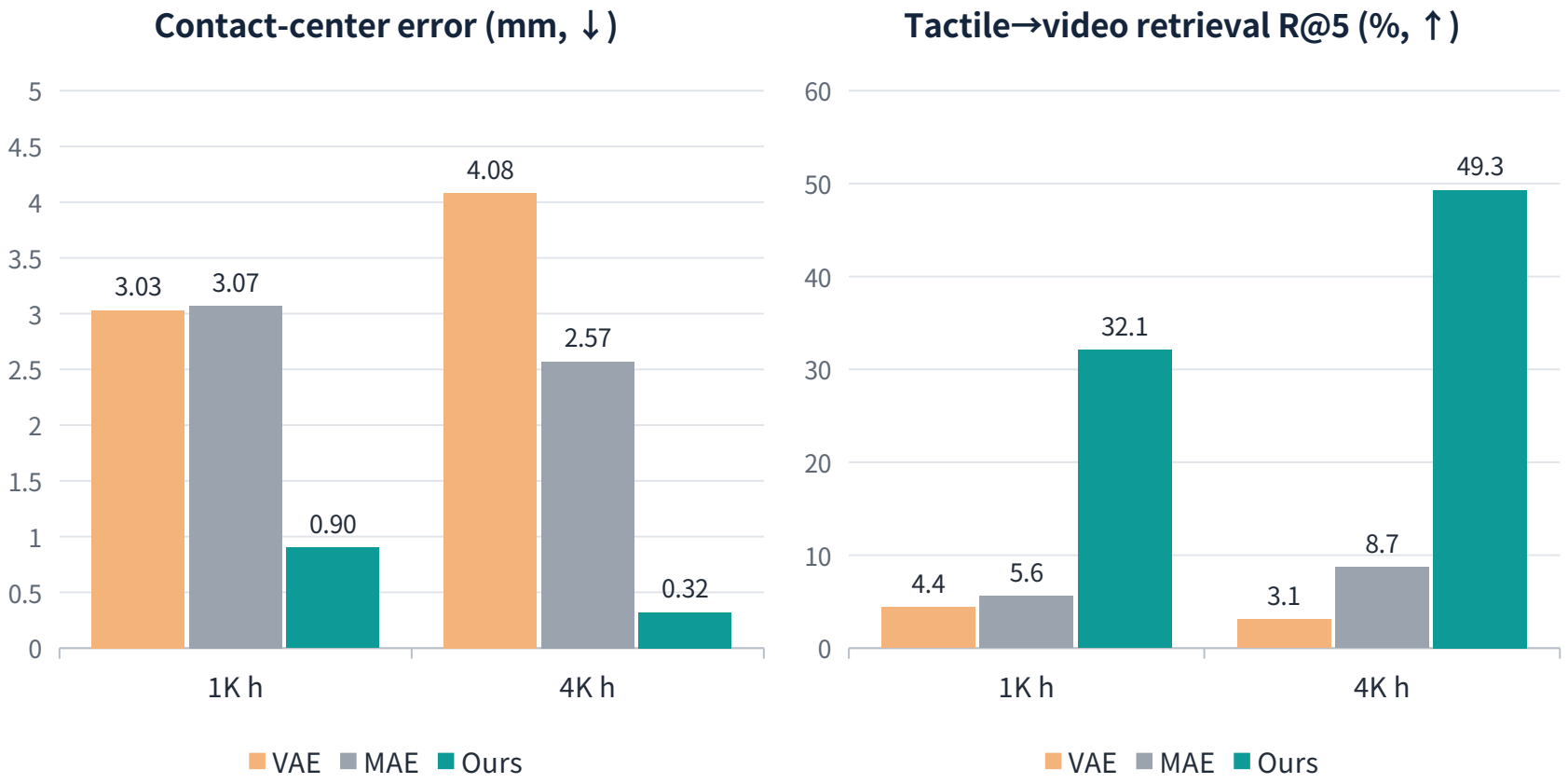
- Magnetic (tested on XHand1): force 4.34 → 2.52 N; future force 3.77 → 1.84 N
- Visuotactile: contact center 0.41 → 0.68 mm (only regression)

Fixed 1,000 h budget → single-source wins 7/8

- Contact center: visuotactile 0.34 vs 0.68 mm; magnetic 0.33 vs 1.12 mm
- Small gaps in temporal prediction; multi-source even better for visuotactile future force (1.49 vs 1.67 N)

Takeaway: scale data with the "easiest-to-collect" sensor and deploy on any hand, at the cost of some spatial precision.

Latent prediction beats reconstruction: VAE even worsens with more data



LingBot-VLA • 5 real tasks

No touch	55.9%
+ VAE (4K)	≈65%
+ MAE (4K)	≈62%
+ Ours (4K)	≈81%

VAE / MAE fall below the no-touch baseline on cup unstacking and plate wiping; Ours is best on all 5 tasks.

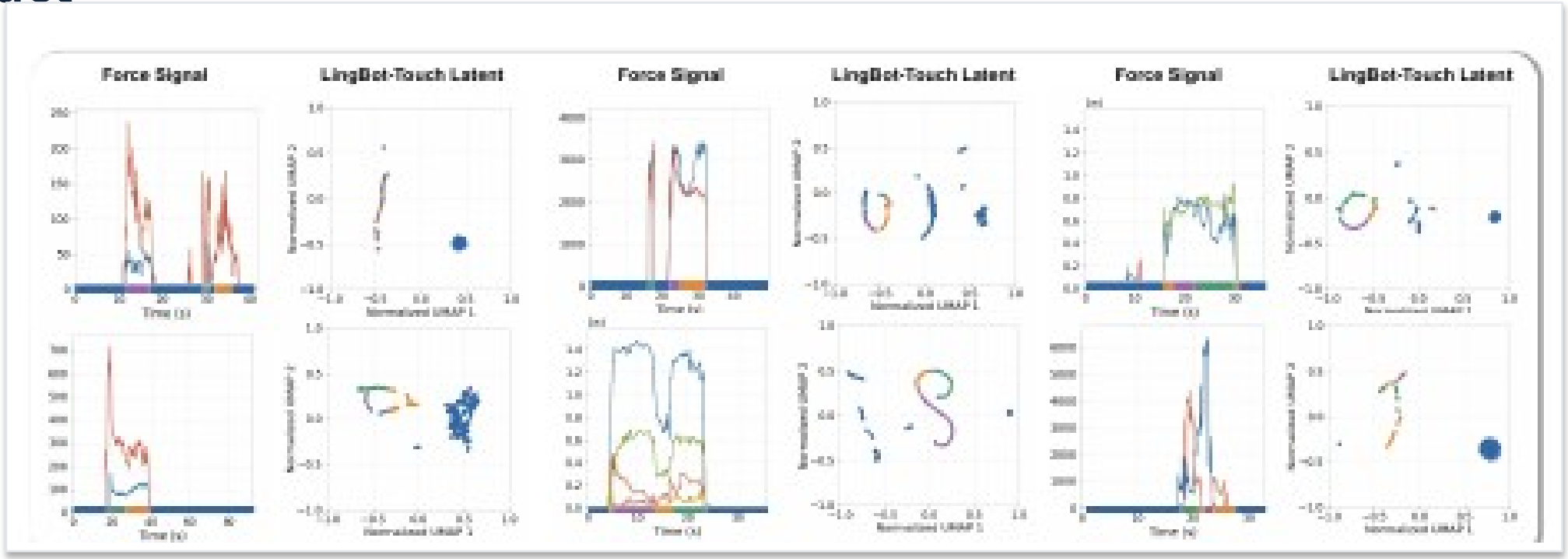
All three have the same params (5.46M); MAE and Ours both use 80% masking. Real-robot means are approximations computed from Fig.9 bar readings (baseline 55.9%, consistent with Fig.13).

Ablations: mask ratio, EMA rate, augmentation, grouped interfaces ~~decide~~

Design axis	Best setting	Key evidence (Sharpa Wave, frozen probing)
Spatial mask ratio	**80%**	Best on all 5 metrics; 90% → retrieval R@5 only 14.4%, 80% → 35.4%
Temporal mask span	**Whole clip**	[100,100]% best on 4/5: no "copying" masked taxels from adjacent frames
Data augmentation	**Value + position**	Retrieval 25.6% (no aug.) → 37.1%; contact center 0.64 → 0.24 mm
clip / tubelet	**32 frames • 2–4 frames/token**	≈1.07 s @30Hz; 1- or 8-frame tubelets clearly worse
EMA momentum	**0.999 → 1.0**	0.998 → 1.0 collapses: force MAE > 7 N, retrieval ≈ 1%
Tokenizer / predictor	**Grouped / group-routed**	Fully shared → grouped: contact center 2.32 → 0.70 mm, best on all 5
Training objective	**target + context**	Adding context loss is roughly neutral (3 slightly better, 2 slightly worse)

Note: all ablations use Tiny, a 1K h subset, 100 epochs; default config (90% masking, etc.) ≠ final config, so compare only within each block.

Latents keep continuous contact evolution, not just “contact / no contact”



No contact → one compact cluster

Near-zero inputs form one cluster, far from other points (as expected).

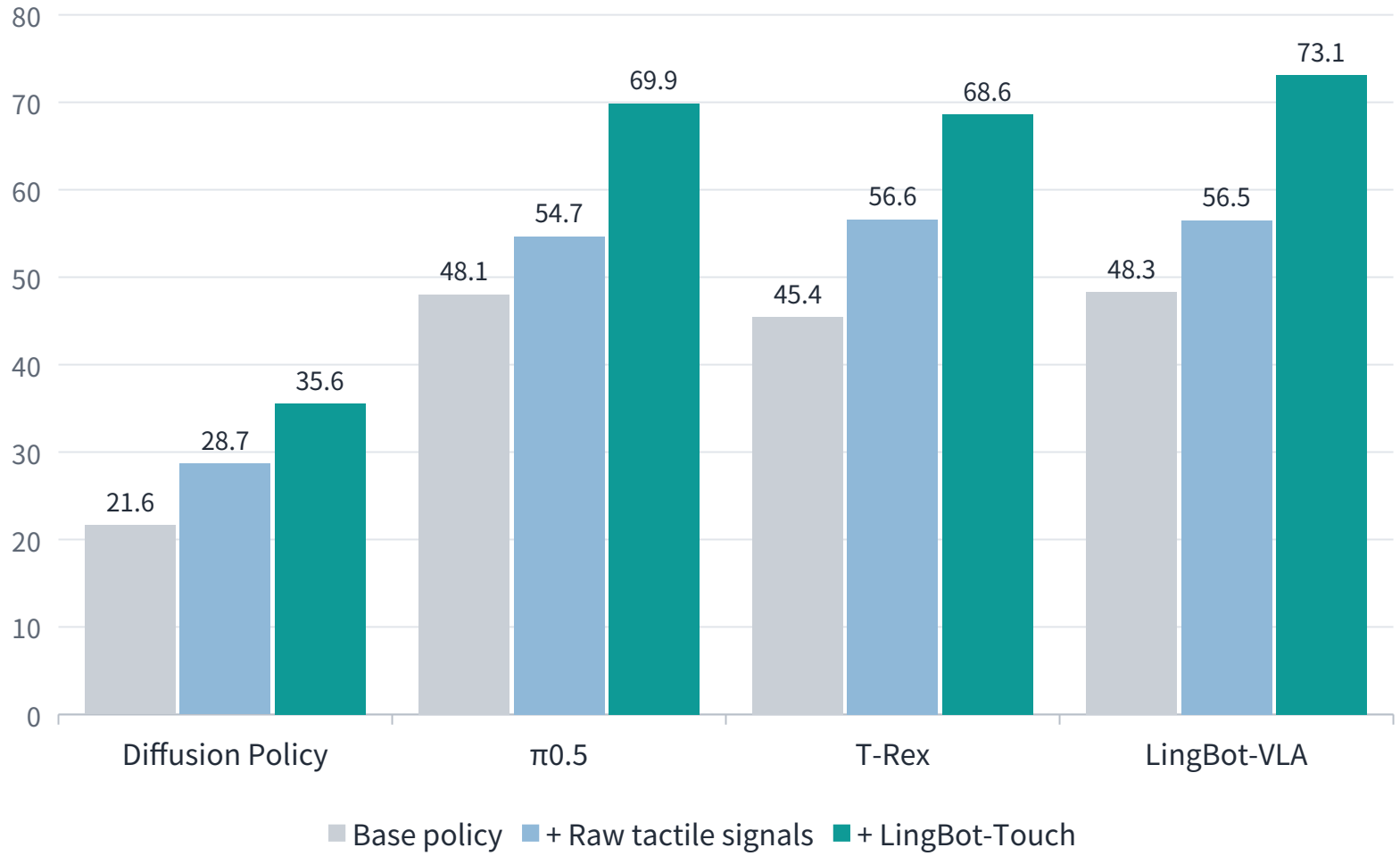
Contact → ordered curve

Force rise/fall/finger shifts: adjacent windows stay close; phases form own arcs.

Implications for policies

Encodes contact “progress”, so policies can gauge grip force and action timing.

Frozen tactile tokens lift 4 policy types by 14–25 pp avg. on 26 real tasks



89 / 104

task×policy combos where Ours is best of three configs; 99 no worse than base policy

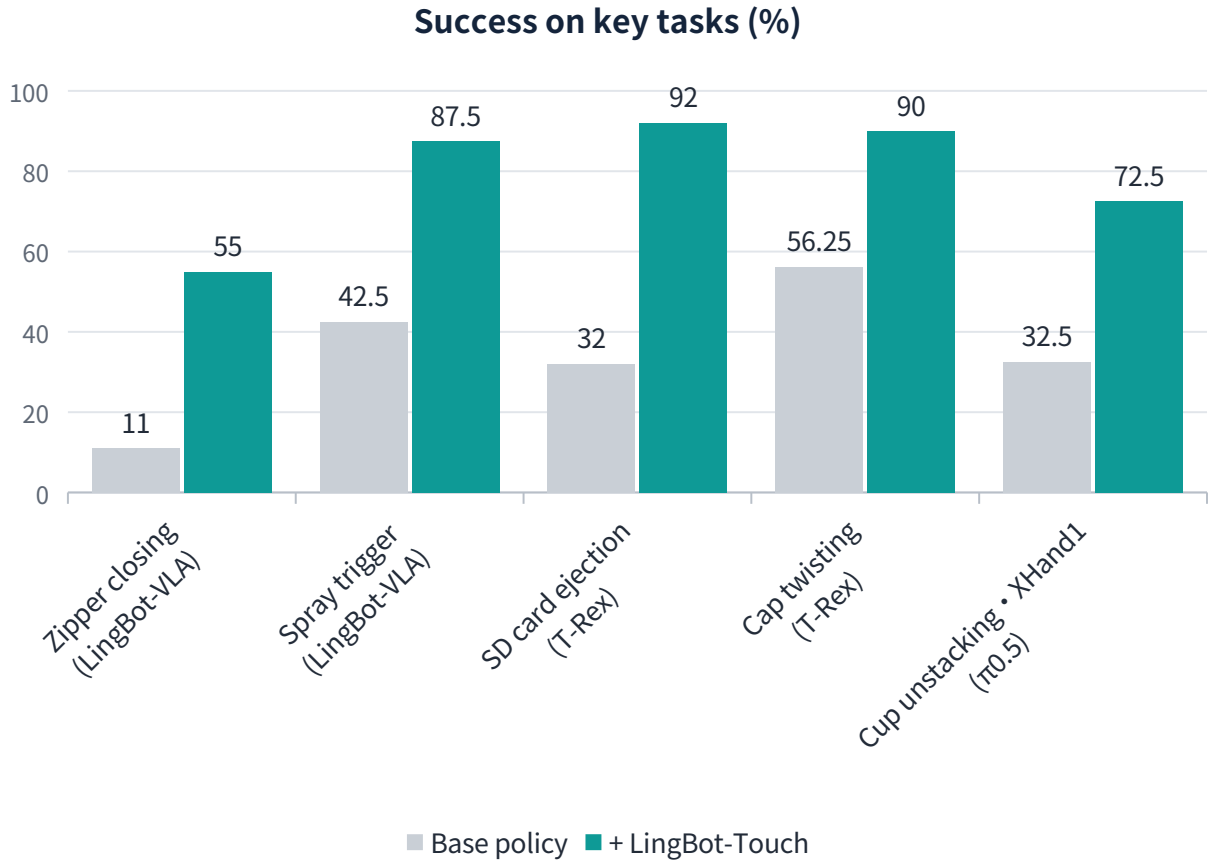
⩽ half

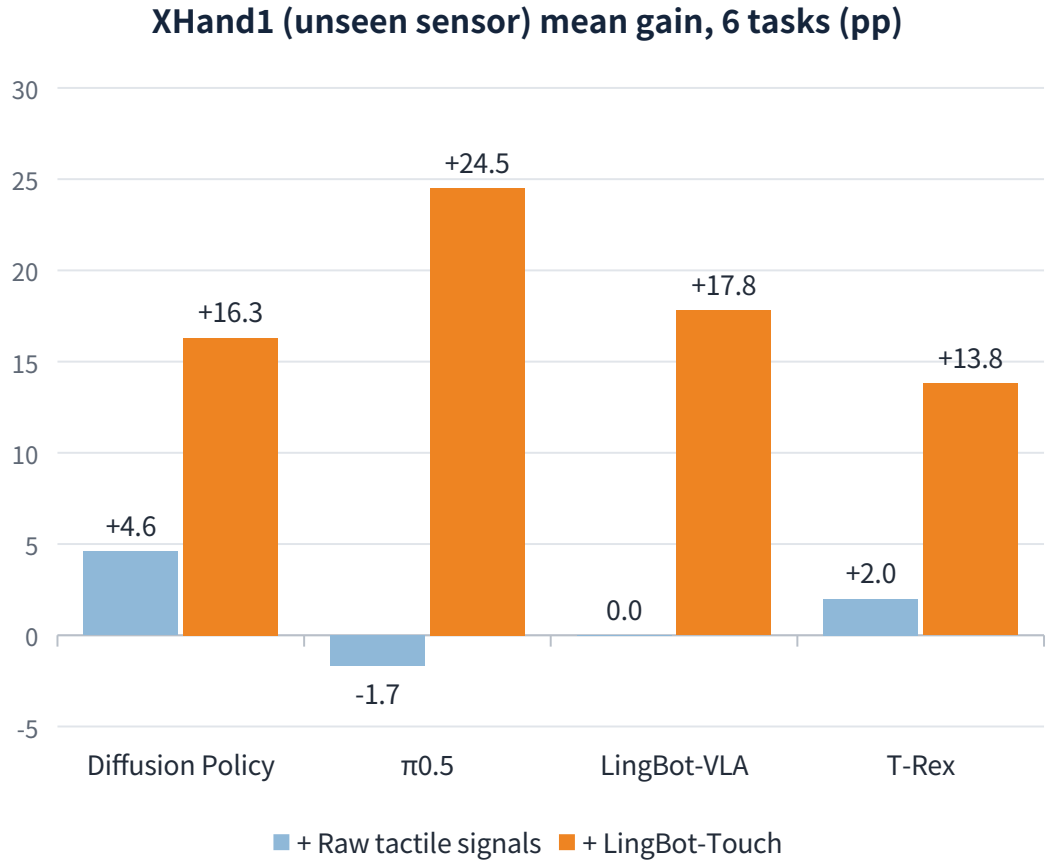
the gain from raw signals (+7.0 / 6.6 / 8.2 / 11.2 pp), and 22/104 combos even get worse

< 0.16%

Encoder share of T-Rex / π0.5 / LingBot-VLA params

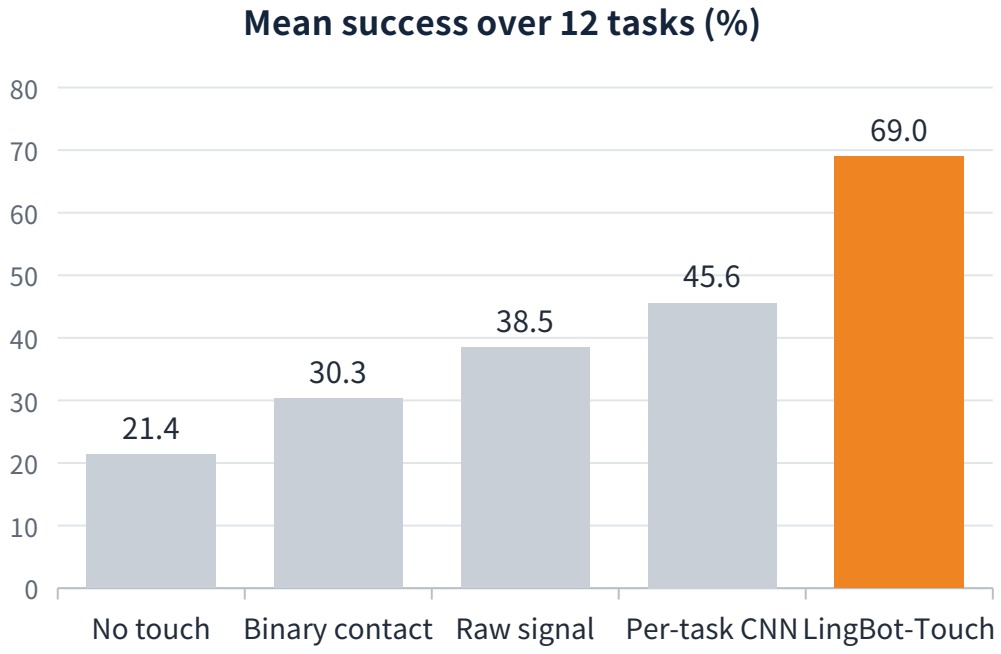
Gains from “knowing touch”: steps vision misses, unseen hands





Not uniformly better: Ours trails the “raw signal” config in 9 of 104 combos (5 of them Diffusion Policy) and the base policy in 5.

Sim RL: pretrained touch, avg. success 21%→69%, stabler seeds

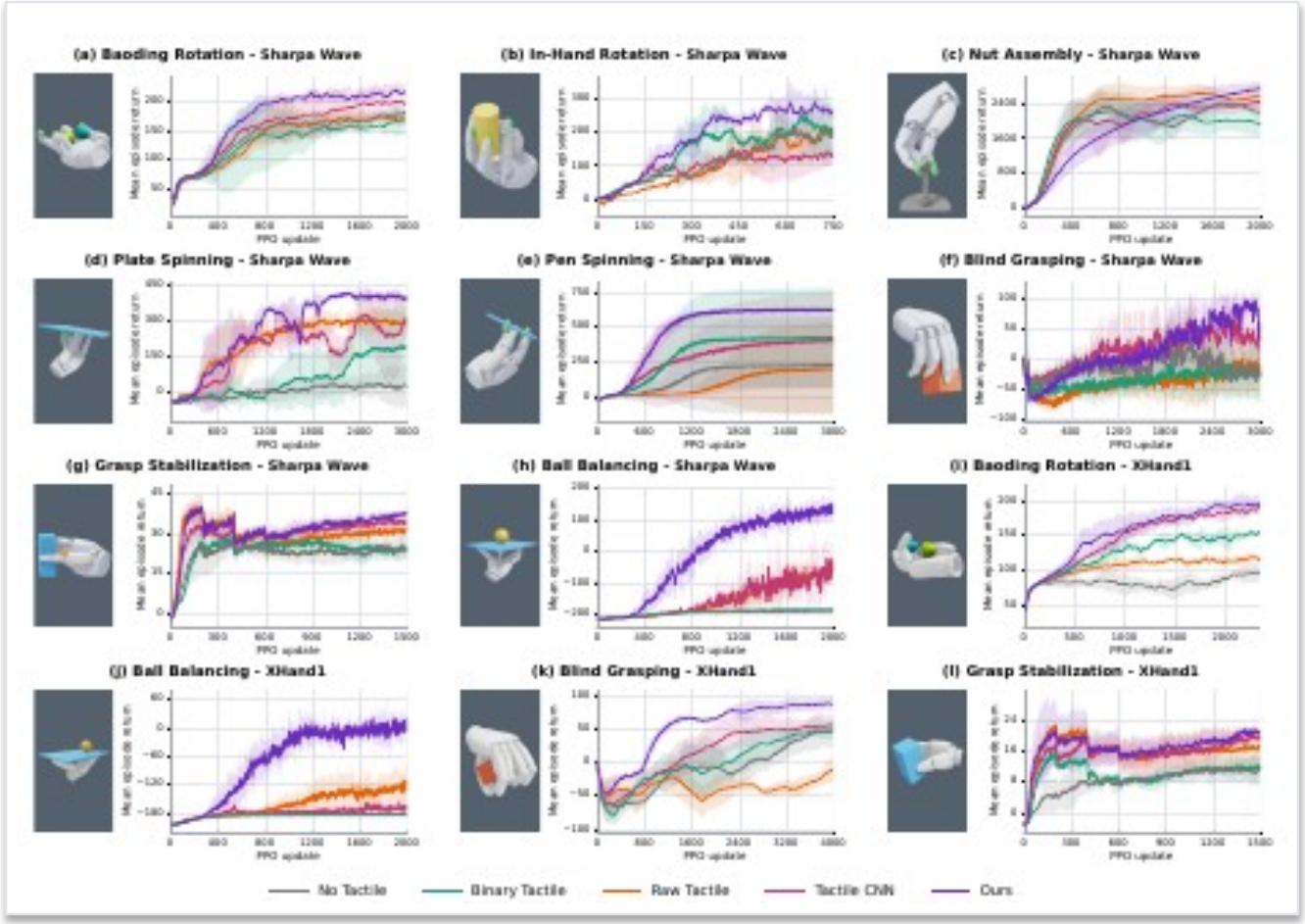


97.1%

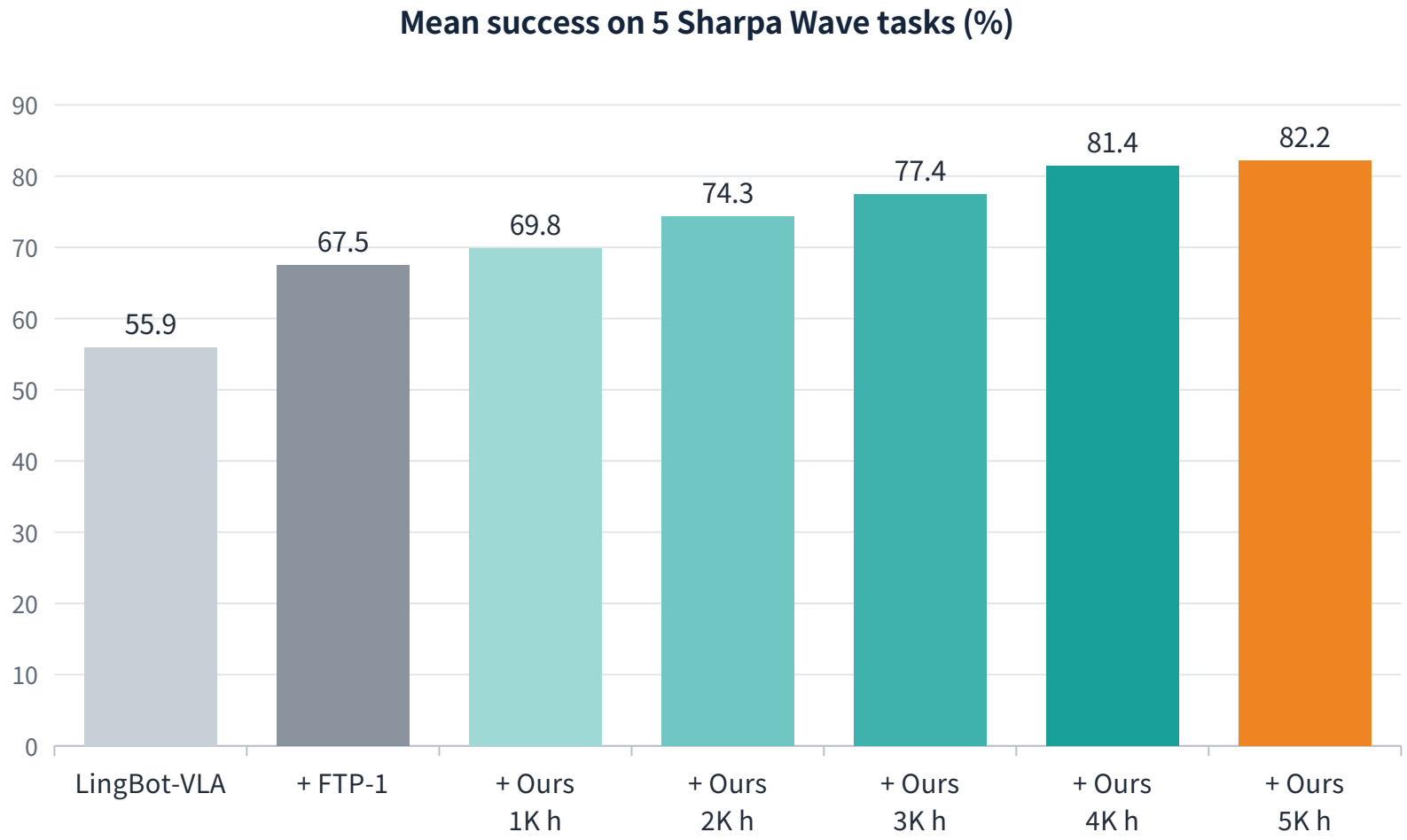
Pen spin (2nd: 64.2%)
Turntable 97.4% (2nd: 82.9%)

σ ⩽ 9.2

Ours' cross-seed std (mean 3.9); others up to 55



Probe rankings carry over to policies: more pretraining data, stronger VLA



+26.3 pp

Gain from the 5K h encoder (55.9% → 82.2%)

Monotonic rise

Every extra 1K h of task-agnostic tactile data improves the policy

67.5%

With FTP-1 encoder: cup unstacking 95→85, plate wiping 74→67, below baseline

Critical View

Caveats: “unseen” = same principle; probe labels from signals

Limitations acknowledged by authors

- Limited hardware: real-robot and sim results need validation on more dexterous hands
- Sensor coverage: capacitive sensors excluded from pretraining due to insufficient data quality
- Fingertips only: palm and other finger regions underrepresented; needs extension to the whole hand
- Under a fixed budget, multi-source mixing loses some spatial precision

Reviewer's concerns

- “Unseen” XHand1 is still magnetic, reusing the magnetic tokenizer and requiring known geometry; transfer to new sensing principles untested
- Contact center / area / force labels derive from sensor readings → probes measure “information retention”, not external physical ground truth
- vs. FTP-1: its adapters are trained from scratch on ~2 h of downstream data, so conditions are not fully matched
- Real robot: 20 trials per task, partial-credit scoring, no confidence intervals; few details on raw-tactile baseline integration
- Inconsistent setups: ablation default ≠ final config; Tab.2 uses the 4K h encoder, Fig.13 the 5K h one
- Release has only inference code, a few samples and Tiny/Small weights; pretraining is hard to reproduce

Summary

Contact, not sensors • representation gains • cross-sensor scale

01

Representations belong to contact, not sensors

Canonical fingertip space + 5D RoPE + layout augmentation → one encoder serves unseen hands with zero parameter updates and beats FTP-1 with dedicated adapters across the board.

02

Gains come from representation, not modality

Frozen tokens improve IL, VLA and RL alike; concatenating raw tactile signals yields at most half the gain and often hurts → touch should enter policies as pretrained semantic tokens.

03

Scale accumulates across sensors

Force and dynamics transfer across principles: scale data with the easiest-to-collect gloves, deploy on any hand; latent prediction keeps improving with data, while reconstruction stalls or degrades.